

US IT Recruitment Analytics

End-to-End Data Analyst Portfolio Project

SQL • Python • Excel • Power BI

1. Business Problem

TechStaff Solutions lacks a centralized analytical view of its US IT recruitment operations. Management needs to understand recruitment demand, candidate application patterns, sourcing-channel performance, offer acceptance, compensation, and recruitment process efficiency.

The organization has recruitment data across candidates, jobs, applications, interviews, offers, and companies. Without structured analysis and visualization, it is difficult to identify recruitment bottlenecks, understand hiring demand, evaluate sourcing channels, and make data-driven decisions.

Project goal: Analyze recruitment data using SQL, Python, Excel, and Power BI to identify actionable insights and recommend improvements to recruitment effectiveness.

2. Business Objectives

- Analyze recruitment demand by role family, job level, and work arrangement.
- Evaluate candidate application patterns and candidate interest across recruitment categories.
- Analyze recruitment source performance and offer outcomes.
- Evaluate offer volume, acceptance rate, and compensation patterns.
- Measure recruitment process efficiency using application-to-offer time.
- Identify recruitment bottlenecks and opportunities.
- Provide data-driven recommendations to improve recruitment effectiveness.

3. Stakeholders

Stakeholder	Primary information needed
Recruitment Manager	Overall recruitment performance, funnel efficiency, offer acceptance, sourcing performance
Recruiting Team Lead	Candidate pipeline, sourcing channels, role demand, bottlenecks
Recruiters	Candidate demand, job requirements, sourcing effectiveness, offer outcomes
Operations Manager	Process efficiency, workload, operational performance
Business Development / Account M	Hiring demand, job requirements, client performance
Senior Management	Hiring trends, compensation, efficiency, strategic recommendations

Data limitation: The dataset does not contain recruiter-level identifiers, so individual recruiter performance was not analyzed.

4. Business Questions

- Which job role families have the highest hiring demand?
- Which job levels receive the most candidate applications?
- Which recruitment sources generate the highest application volume?
- Which recruitment sources have the highest offer acceptance rates?
- What is the average offer salary across different job levels?

- Which work arrangements attract the most applications?
- How does candidate experience vary across the candidate pool?
- How many applications are generated per available job across role families?
- What is the average time from application to offer?
- What major recruitment bottlenecks and opportunities can be identified?

5. KPIs & Metrics

KPI	Definition	Value
Total Candidates	Unique candidates	9,000
Total Jobs	Total job postings/requisitions	2,000
Total Applications	Total candidate applications	15,000
Total Interviews	Recorded interviews	7,196
Total Offers	Offers generated	938
Accepted Offers	Offers accepted	699
Offer Acceptance Rate	Accepted offers / total offers	74.52%
Average Offer Salary	Average offered base salary	\$129,879.69
Application-to-Offer Time	Average days from application to offer	28.70 days
Applications per Job	Applications / jobs	7.50

Metric note: Application-to-Offer Time is used instead of Time-to-Hire because the dataset does not contain a candidate joining/placement date.

6. Dataset & Data Sources

The project uses a synthetic US IT Job Postings & Applications dataset containing recruitment-related information across candidates, companies, job postings, applications, interviews, and offers. The dataset was selected because its structure represents a realistic recruitment analytics environment.

Table	Records
Companies	320
Candidates	9,000
Jobs	2,000
Applications	15,000
Interviews	7,196
Offers	938

Data source: Hugging Face synthetic, privacy-safe recruitment dataset by mindweave. The analysis demonstrates a realistic analytics workflow and does not represent actual company performance.

Tools: Excel for inspection/cleaning/validation; Python for cleaning and EDA; MySQL for relational modeling and SQL analysis; Power BI for modeling, KPIs, visualization, and dashboarding.

7. Data Quality Audit

Check	Result
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Duplicate primary IDs	No duplicates found
Missing values	Reviewed and handled based on business meaning

Check	Result
Data types	Checked and standardized
Referential integrity	Related-table relationships validated
Salary values	Offer salaries checked for positive values
Offer acceptance dates	Validated against acceptance status
Date consistency	Date fields reviewed and standardized
Categorical values	Reviewed for consistency
Business rules	Key recruitment rules checked

- Primary identifiers across the six tables contained no duplicate records.
- Relationships between candidates, jobs, applications, interviews, and offers were validated.
- Null accepted_at values were retained for offers that were not accepted rather than artificially filled.
- An acceptance_status field was created to distinguish Accepted and Not Accepted offers.
- Boolean acceptance values were standardized to 1 = Accepted and 0 = Not Accepted.

8. Data Cleaning & Transformation

- Standardized column names and data formats.
- Converted relevant date fields to datetime format.
- Handled missing values according to business meaning.
- Created acceptance_status from accepted_at.
- Standardized accepted to 1/0 for database analysis.
- Created date fields including date, month, year, and month-year.
- Created experience buckets: 0–2, 3–5, 6–10, and 10+ years.
- Created salary_midpoint and salary_range for job salary analysis.
- Validated cleaned datasets before loading them into MySQL and Power BI.

Cleaned files: companies_clean.csv, candidates_clean.csv, jobs_clean.csv, applications_clean.csv, interviews_clean.csv, offers_clean.csv.

9. Data Model & Relationships

The project uses a relational recruitment model with six core entities.

- companies many jobs
- candidates many applications
- jobs many applications
- applications many interviews
- applications many offers

The model supports analysis across candidate, job, company, application, interview, and offer dimensions while maintaining consistent identifiers.

10. SQL Analysis

MySQL was used to analyze recruitment volume, sourcing performance, client demand, job characteristics, compensation, and process efficiency.

Analysis area	Key result
Current application stage	Applied 8,638; Screened 2,792; Interview 2,632; Offer 938
Stage shares	Applied 57.59%; Screened 18.61%; Interview 17.55%; Offer 6.25%
Interview Offer	26.27%
Offer Acceptance	74.52%
Top source by applications	Job board: 7,094 applications
Highest offer acceptance by source	Recruiter: 82.14%
Average offer salary	\$129,879.69
Highest average salary	Lead: \$174,179.50
Highest job demand	Operations: 991 jobs
Average application → offer time	28.70 days

Interpretation note: applications.stage represents the current stage in the dataset, not a complete historical stage-transition log. It should therefore not be described as a true historical funnel without stage-history data.

11. Python EDA

Python was used for univariate, bivariate, and multivariate exploratory data analysis. The EDA validated patterns identified through SQL and explored relationships among experience, application score, job characteristics, sourcing channels, salary, and recruitment outcomes.

- Experience level showed relatively modest differences in recruitment conversion.
- Application score showed a limited relationship with offer conversion.
- Offer salary increased substantially with job level.
- Recruiter sourcing produced the strongest offer acceptance rate.
- Operations represented the largest job-demand segment.
- Average application-to-offer time was approximately 28.7 days.
- Application volume declined after August 2025; February 2026 appears to be a partial month and should not be compared directly with full months.

12. Power BI Dashboard

The Power BI report was organized into four pages to provide an executive-to-detail analytical flow.

Page	Purpose
1. Recruitment Overview	Executive KPIs, overall recruitment picture, source and role demand
2. Recruitment Performance	Offer acceptance, salary, source, level, role, and location performance
3. Job & Candidate Analysis	Job demand, applications per job, experience, level, location, employment type
4. Executive Insights & Recommendations	Key findings, business implications, and recommended actions

13. Key Insights

- Only 6.25% of current application records are at the offer stage, highlighting substantial drop-off before offers.
- The overall offer acceptance rate is approximately 74.5%, indicating relatively strong offer conversion.
- Recruiter-sourced offers have the highest acceptance rate at approximately 82.1%, while LinkedIn is lowest at approximately 67.9% among analyzed sources.
- Operations accounts for approximately 49.6% of job postings, making it the largest hiring-demand segment.
- Lead roles have the highest average offer salary at approximately \$174K, followed by Senior, Mid, and Junior levels.
- Remote roles receive the highest application volume in the analyzed dataset, followed by hybrid and onsite roles.
- Product has the highest applications-per-job ratio at approximately 7.81, followed by Data at approximately 7.59.

14. Business Recommendations

- Review screening and candidate-selection practices to identify causes of application-to-offer drop-off.
- Increase focus on recruiter and referral sourcing while investigating lower acceptance performance from LinkedIn candidates.
- Align recruitment capacity with role demand, especially the high-volume Operations segment.
- Maintain specialized pipelines for technical roles such as Engineering and Data.
- Use salary benchmarks by job level to support competitive and consistent offers.
- Monitor application-to-offer time regularly and investigate process delays.
- Use applications-per-job as a demand-intensity indicator when prioritizing sourcing and recruiting resources.

15. Limitations

- The dataset is synthetic and does not represent actual company recruitment performance.
- Recruiter-level identifiers are not available, preventing individual recruiter productivity analysis.
- There is no candidate joining/placement date, so true Time-to-Hire cannot be calculated.

- The application stage field represents current stage rather than a full historical stage-transition log.
- Revenue, client billing, placement fees, and recruiter cost data are not available.
- Some date periods may be partial, particularly February 2026, so they should be interpreted carefully.

16. Project Outcome

The project demonstrates an end-to-end data analytics workflow: business understanding, stakeholder identification, KPI definition, data quality auditing, cleaning and transformation, relational data modeling, SQL analysis, Python EDA, Power BI visualization, insight generation, and business recommendations.

The final dashboard provides management with a structured view of recruitment demand, candidate behavior, sourcing performance, offer outcomes, compensation, and process efficiency.